

Nutrition and Dietary Health — A Clinical Guide

1. Fundamentals of Nutrition

Nutrition is the process by which living organisms obtain and use food substances for growth, energy, and maintenance of bodily functions. Good nutrition is fundamental to health, disease prevention, and recovery from illness.

Macronutrients: Carbohydrates (4 kcal/g) — primary energy source. Proteins (4 kcal/g) — growth, repair, enzymes, antibodies. Fats (9 kcal/g) — energy storage, fat-soluble vitamins, cell membrane integrity, hormones. Water — not caloric but essential for all metabolic processes.

Micronutrients: vitamins (fat-soluble: A, D, E, K; water-soluble: B complex, C) and minerals (macrominerals: calcium, phosphorus, magnesium, sodium, potassium; trace elements: iron, zinc, iodine, copper, selenium, fluoride) required in small amounts but essential for specific physiological functions.

2. Protein-Energy Malnutrition in India

India has one of the world's highest burdens of child malnutrition. NFHS-5 (2019-21): 35.5% of children under 5 are stunted (low height-for-age), 19.3% are wasted (low weight-for-height), 32.1% are underweight. Despite progress, India accounts for ~30% of global stunted children.

Protein-Energy Malnutrition (PEM): Severe Acute Malnutrition (SAM) — weight-for-height < -3 SD (Z-score) or MUAC < 11.5 cm. Marasmus: severe wasting, 'skin and bones', no oedema. Kwashiorkor: oedema (bilateral pitting), hypoalbuminaemia, dermatosis, 'moon face', hair changes (flag sign). Marasmic-Kwashiorkor: both features.

Management of SAM: 10-step WHO protocol in NRCs (Nutritional Rehabilitation Centres). Phase 1 (medical stabilisation): treat hypoglycaemia (10% glucose), hypothermia, dehydration (with ReSoMal), infections (amoxicillin empirically), micronutrients (not iron yet — infections present). Phase 2 (rehabilitation): RUTF (Ready-to-Use Therapeutic Food — F100, Plumpy'Nut) for catch-up growth.

3. Obesity and Overweight

Overweight (BMI 25-29.9 kg/m²) and obesity (BMI ≥30 kg/m²) are rapidly increasing in India — 24% of urban adults are overweight/obese (NFHS-5). South Asian-specific cut-offs: overweight ≥23 kg/m²; obese ≥27.5 kg/m².

Obesity is associated with: Type 2 diabetes, hypertension, dyslipidaemia (metabolic syndrome), cardiovascular disease, NAFLD, sleep apnoea, osteoarthritis, GERD, infertility (PCOS), depression, and increased cancer risk (breast, colon, endometrial, oesophageal, pancreatic, kidney, gallbladder).

Management: dietary modification (energy deficit 500-1000 kcal/day below calculated needs — no single diet has clear superiority; Mediterranean, DASH, and low glycaemic index diets all effective), physical activity (150-300 min/week moderate intensity), behavioural therapy.

Pharmacotherapy (BMI ≥30 or ≥27.5 with comorbidities): orlistat (lipase inhibitor), naltrexone-bupropion, phentermine-topiramate, liraglutide 3 mg (Saxenda), semaglutide 2.4 mg weekly (Wegovy — 15-17% weight loss). Metabolic/bariatric surgery (BMI ≥40 or ≥35 with comorbidities): sleeve gastrectomy, Roux-en-Y gastric bypass — most effective long-term intervention.

4. India-Specific Nutritional Concerns

Vitamin B12 deficiency is extremely common in India — particularly among vegetarians/vegans (B12 found only in animal products). Estimates suggest 47-80% of urban Indians are deficient. Often subclinical but causes progressive neurological damage. All strict vegetarians should supplement B12.

Vitamin D deficiency: paradoxically very common in sunny India — estimated 50-90% of Indians are deficient. Reasons: dark skin pigmentation (higher melanin reduces cutaneous synthesis), indoor lifestyles (urban), modest-covering clothing, vegetarian diet (few natural food sources). All adults should supplement 1000-2000 IU Vitamin D3 daily.

Iron deficiency: most common micronutrient deficiency globally and in India (see anaemia chapter). Bioavailability of iron from plant sources (non-haem iron) is much lower (2-5%) than haem iron from meat (15-35%). Phytates in cereals and polyphenols in tea significantly reduce iron absorption.

Calcium: majority of Indians consume well below the recommended 1000-1200 mg/day. Milk consumption is a major source but many are lactose intolerant. Plant sources: finger millet (ragi, 344 mg/100g), amaranth, sesame seeds, green leafy vegetables. Vitamin D is essential for calcium absorption.

5. Therapeutic Diets

DASH Diet (Dietary Approaches to Stop Hypertension): high in fruits, vegetables (8-10 servings/day), low-fat dairy, whole grains, legumes, nuts; low in sodium (<2300 mg/day), red meat, sweets, and saturated fats. Reduces systolic BP by 8-14 mmHg.

Diabetic diet (Medical Nutrition Therapy): plate method — ½ plate non-starchy vegetables, ¼ whole grains/starches, ¼ lean protein. Low glycaemic index foods (oats, dal, barley, most vegetables) preferred. Limit refined carbohydrates, sugar-sweetened beverages, fruit juices. Distribute carbohydrates evenly across meals.

Mediterranean Diet: abundant plant foods, olive oil as primary fat, fish 2x/week, poultry and dairy in moderation, red meat rarely. Rich in omega-3 fatty acids, polyphenols, fibre. Proven to reduce CVD events, T2DM, cognitive decline, and certain cancers.

Low-FODMAP diet: for irritable bowel syndrome (IBS) — restricts fermentable carbohydrates that cause bloating, gas, diarrhoea, and constipation. Eliminates high-FODMAP foods (onion, garlic, wheat, lactose, legumes, certain fruits) for 6-8 weeks then systematically reintroduces.

6. Dietary Guidelines for India

ICMR-NIN (National Institute of Nutrition) Dietary Guidelines for Indians 2024 recommend: Consume a balanced diet including at least 400-500g/day of fruits and vegetables. Include whole grains, legumes, and millets (bajra, jowar, ragi, sorghum).

Key recommendations: Reduce ultra-processed foods (UPF), sugar, salt, and saturated/trans fats. Cook with less oil (preferred: groundnut, sesame, mustard, coconut in traditional cuisines; avoid reheating oils). Avoid tobacco and limit alcohol.

Indian plate model (ICMR): cereals/millets 40-45%, pulses 15%, vegetables 30%, dairy/eggs/fish/poultry 10%, fats/oils/sugars <5%.

Millets promotion: India has declared 2023 the International Year of Millets. Millets are nutritionally superior to polished rice and refined wheat — high fibre, iron, calcium, magnesium, B vitamins. Government initiatives (millet missions, Nutri-Cereals) aim to increase production and consumption.

7. Infant and Child Nutrition

Exclusive breastfeeding (EBF) for the first 6 months of life is the single most powerful intervention for infant survival. Breast milk provides ideal nutrition, passive immunity (IgA, lactoferrin), and benefits for mother (reduced breast cancer, postpartum depression). WHO recommends continued breastfeeding up to 2 years alongside complementary foods.

Complementary feeding: introduce from 6 months. Start with iron-rich foods (mashed dal, mashed fortified cereal), then puréed fruits/vegetables, soft family foods. Responsive feeding practices, adequate frequency (2-3 times daily 6-8 months, 3-4 times 9-23 months), and dietary diversity (5+ food groups).

India's Integrated Child Development Services (ICDS): provides supplementary nutrition (take-home rations, hot cooked meals), immunisation, health check-ups, referral services through Anganwadi centres. Mid-Day Meal Programme provides one-third daily nutritional requirement for school children.

8. Nutrition in Chronic Disease Management

Cancer nutrition: malnutrition is common in cancer (cancer cachexia — profound weight loss, muscle wasting). Malnutrition worsens chemotherapy toxicity and reduces survival. Nutritional screening (NRS-2002 or MUST) mandatory at cancer diagnosis. High protein intake (1.2-1.5 g/kg/day). Omega-3 fatty acids (EPA/DHA) may attenuate cachexia.

Renal diet: CKD Stages 3-5 — restrict dietary potassium (avoid bananas, potatoes, tomatoes, oranges), phosphate (reduce dairy, processed foods, cola), and protein (0.6-0.8 g/kg/day pre-dialysis). On dialysis: higher protein required (1.2-1.4 g/kg/day), fluid restriction.

Post-bariatric surgery: risk of micronutrient deficiencies (B12, iron, calcium, Vitamin D, thiamine, zinc, folate) due to malabsorption. Lifelong supplementation with multivitamin, calcium + Vitamin D, B12 mandatory.

9. Food Safety in India

Food adulteration is a significant public health concern in India. Common adulterants: milk diluted with water or added with detergent, spices (chilli powder, turmeric) adulterated with brick dust, artificial colours, lead chromate; edible oils mixed with cheaper or non-food oils; pulses coated with metanil yellow (toxic azo dye).

FSSAI (Food Safety and Standards Authority of India): regulates food safety standards, labelling, imports, and enforcement. FSSAI has introduced Star Health Rating system for packaged food, Front-of-Pack Labelling (FoPL) regulations mandating disclosure of high sugar, salt, and saturated fat content.

Food poisoning: Salmonella, Staphylococcus aureus (pre-formed toxin), E. coli, Bacillus cereus, Clostridium perfringens, Listeria are common pathogens. Food safety practices: wash hands, cook food thoroughly ($\geq 70^{\circ}\text{C}$ core temperature), refrigerate promptly, separate raw/cooked foods, use safe water.

10. Nutrition Programmes in India

POSHAN Abhiyaan (2018): National Nutrition Mission targeting reduction of stunting, wasting, anaemia, and low birth weight. Phase 1 focused on 315 districts with high malnutrition. Real-time monitoring through POSHAN Tracker app.

PM POSHAN Yojana (formerly Mid-Day Meal scheme): hot cooked nutritious meal for students (Classes 1-8) in government and government-aided schools. Reaches approximately 120 million children. Reduces dropout rates and improves learning outcomes alongside nutrition.

National Food Security Act (NFSA) 2013: legal right to food for 75% of rural and 50% of urban population — subsidised rice (Rs 3/kg), wheat (Rs 2/kg), coarse grains (Rs 1/kg) through Public Distribution System (PDS).

One Rank, One Pension — ICDS and POSHAN combined have contributed to significant reduction in under-5 stunting (41.5% in 2014 to 35.5% in 2020, NFHS data). India's progress has been mixed — economic growth has reduced poverty-related malnutrition but dietary diversity improvements have been slow.

